

Predicting Progression to Chronic Pancreatitis Using Multimodal Time-to-Event Deep Learning Models

Yicen Yang

yy3539@nyu.edu

Haojie Yin

hy2108@nyu.edu

Yujia Zhang

yz6553@nyu.edu

Haoying Xu

hx2504@nyu.edu

Abstract

Chronic Pancreatitis (CP) can arise after Acute Pancreatitis (AP), motivating early prediction of patients at elevated risk. We propose a multimodal time-to-event framework that integrates abdominal CT imaging with structured clinical data to estimate CP progression risk. Given the limited size of our CT-imaging subset, we adopt a SAM-Med2D pretrained encoder to obtain slice-level medical features, which are aggregated into volume-level representations and fused with clinical embeddings. The model is trained with a Cox partial-likelihood objective using a one-year predictive horizon. Results show that our multimodal approach outperforms clinical-only and imaging-only baselines, demonstrating the benefit of combining foundation-model imaging features with survival modeling for CP risk prediction.

1 Introduction

Acute Pancreatitis (AP) can evolve into Chronic Pancreatitis (CP), a severe and irreversible fibro-inflammatory disease associated with long-term morbidity. Predicting the likelihood and timing of CP progression remains a significant clinical challenge due to heterogeneous patient trajectories and the limitations of existing scoring systems.

Recent advances in medical deep learning—particularly volumetric imaging models such as UNETR, which leverages Transformers for 3D anatomical representation learning—have demonstrated strong performance in segmentation and regional feature extraction (Hatamizadeh et al., 2021). Similarly, CT-based deep learning models for pancreatitis diagnosis have shown high accuracy in identifying inflammatory regions and disease severity (Zhang et al., 2024). However, these methods primarily target detection or segmentation rather than *longitudinal risk prediction* or *time-to-event modeling*.

At the same time, survival analysis models such

as DeepSurv (Katzman et al., 2018) have proven effective for structured clinical data but lack the capacity to incorporate rich spatial information from medical imaging. This creates a methodological gap: how to jointly model CT imaging and clinical metadata in a unified deep survival framework for CP progression prediction.

To address this challenge, we initially considered using a Vision Transformer (ViT)–based encoder for CT representation learning, motivated by ViT’s strong ability to capture long-range dependencies in medical images (Dosovitskiy et al., 2021). However, ViT models typically require large-scale datasets for stable training, and our CT-imaging subset—approximately 2,300 abdominal studies—is too small to support ViT training from scratch without severe overfitting, despite the larger clinical cohort.

To mitigate overfitting and leverage strong pretrained representations, we adopt a SAM-Med2D pretrained 2D medical image encoder (Wang et al., 2023), which provides slice-level features enriched with medical priors and enables aggregation into 3D or sequence-level embeddings for downstream survival prediction. This approach aligns with recent findings that pretrained foundation models substantially enhance performance in low-data clinical settings. Consequently, SAM-Med2D offers a more robust and data-efficient alternative to our original ViT-based design.

Our work therefore bridges survival modeling and multimodal deep learning by integrating pretrained imaging encoders with structured clinical variables into a unified time-to-event prediction framework for CP progression.

2 Related Work

Deep Learning for Pancreatic Imaging. CT-based pancreatitis detection and lesion segmentation have shown strong performance using CNN

and U-Net architectures. The recent diagnostic model for acute pancreatitis (Zhang et al., 2024) demonstrates that deep neural networks can reliably identify peripancreatic inflammation and necrosis. For more general 3D medical imaging, the UNETR architecture (Hatamizadeh et al., 2021) introduces a volumetric Transformer encoder capable of capturing long-range dependencies, which motivates our use of a Transformer backbone for CT-based risk modeling.

Pretrained Medical Image Encoders. Recent progress in foundation models for medical imaging has produced large-scale pretrained encoders that substantially improve data efficiency. A notable example is SAM-Med2D (Wang et al., 2023), a medical-domain adaptation of the Segment Anything Model (SAM) that is trained on millions of 2D radiology images. SAM-Med2D provides strong universal feature representations for CT, MRI, and ultrasound, enabling downstream tasks to achieve high performance even with limited training data. Given that our dataset contains only approximately 2,000 CT studies, training a vision backbone entirely from scratch would likely lead to severe overfitting.

Vision Transformers for Medical Imaging. Vision Transformers (ViT) (Dosovitskiy et al., 2021) have demonstrated strong performance in both natural and medical image analysis due to their ability to capture long-range dependencies without relying on convolutional inductive biases. ViT-based medical models—including UNETR for 3D segmentation—have shown promising results in volumetric CT and MRI analysis. Our initial plan was to adopt a ViT-style encoder for feature extraction. However, ViT architectures typically require substantially larger datasets for stable training and generalization. Considering our relatively small dataset size, we decided *not* to train a ViT from scratch.

Survival Models and Time-to-Event Prediction. Classical proportional hazards models, such as Cox regression, have been extended to deep learning via DeepSurv (Katzman et al., 2018). These models learn nonlinear mappings from covariates to hazard functions and often outperform traditional survival models in clinical datasets. However, they are typically limited to structured EHR variables. Our work integrates DeepSurv-inspired survival loss into a multimodal architecture that incorporates

imaging and metadata.

3 Dataset and Method Overview

3.1 Dataset

Our study uses a multimodal dataset consisting of (1) abdominal CT imaging collected at the time of Acute Pancreatitis (AP) presentation, and (2) structured clinical metadata aggregated from electronic medical records. Each modality provides complementary information relevant to predicting long-term progression to Chronic Pancreatitis (CP).

Imaging Dataset. The raw imaging source consists of 43,161 tar archives, each containing a full CT study composed of sequential DICOM slices. In total, the dataset includes more than 42.8 million DICOM files. Each DICOM contains both pixel data and rich acquisition metadata. As described in our preprocessing pipeline, we processed the dataset in parallel by dividing it into 309 chunks and extracting metadata and pixel arrays from each slice. All RGB images were converted to grayscale, multi-frame images were reduced to a representative 2D slice, and pixel intensities were standardized to `int16`. To reduce storage from 25 TB to 8 TB, all slice arrays were compressed using BLOSC (LZ4HC + bitshuffle). Because slices vary in height and width, slices were grouped by resolution before compression. The result is a set of 309 metadata CSVs and compressed imaging blobs covering the entire dataset.

These preprocessed slices form the basis for our SAM-Med2D feature extraction: each slice is processed independently through a pretrained 2D encoder, after which slice-level embeddings are aggregated into volume-level representations.

This CT-derived imaging cohort corresponds to 2,300 patients who have both clinical data and at least one abdominal CT study, forming the subset used for multimodal survival modeling.

Clinical Dataset. The clinical dataset contains 4,904 patients with structured electronic medical record (EMR) data, including demographics, laboratory biomarkers, anthropometrics, diagnoses, encounter history, and behavioral risk indicators. Each patient has two identifiers: MRN for the medical imaging data and `pat_id` for the clinical data. We normalize MRN to ensure consistent linkage between imaging records and the corresponding clinical `pat_id`. For imaging-clinical alignment, we restrict the cohort to patients who also have

at least one valid abdominal CT study, yielding a CT-matched subset of 2,300 patients. This filtering affects only multimodal experiments; all clinical-only baselines (e.g., DeepSurv) are trained on the full 4,904-patient cohort. After preprocessing, each patient contributes one row of structured clinical covariates, while CT-matched patients additionally contribute a variable-length sequence of abdominal slices for multimodal modeling.

Survival Label Preprocessing. Each patient’s time-to-event label is defined as the interval between the earliest Acute Pancreatitis (AP) encounter and the first recorded diagnosis of Chronic Pancreatitis (CP). The raw duration, originally in days, is converted to years. Patients who progress to CP receive an event indicator $\delta = 1$, whereas those without progression are right-censored ($\delta = 0$) at their last follow-up. This raw survival representation provides a unified foundation for downstream experimental label constructions and is shared across both clinical and imaging models.

3.2 Method Overview

Our complete modeling pipeline is implemented in `pipeline_main_v2.2.ipynb`. The model integrates 2D slice-level encoders, volumetric feature aggregation, clinical metadata modeling, multimodal fusion, and a Cox partial-likelihood objective for time-to-event prediction.

3.3 Imaging Pipeline 1

Each CT study consists of a variable-length sequence of 2D DICOM slices. Our imaging pipeline converts these heterogeneous slices into a fixed-dimensional patient-level representation through four stages: slice grouping and sampling, metadata normalization, pretrained image encoding, and transformer-based aggregation.

Slice Grouping and Sampling. Raw DICOM slices vary widely in number, orientation, and acquisition parameters. We first group slices by `SeriesInstanceUID` and `ImageOrientationPatient`, forming coherent anatomical blocks with consistent spatial orientation. Slices are ordered along the Z-axis within each block. All slices in the same block share the same acquisition time (`StudyTime`), while `StudyTime` may differ across blocks. To obtain a computationally manageable yet anatomically representative subset, we apply a two-stage sampling scheme: (1) proportional

sampling across blocks, and (2) equal-Z stratified sampling within each block. The total number of slices per study is capped by a predefined `MAX_TOTAL_SLICES`, ensuring consistent batch construction.

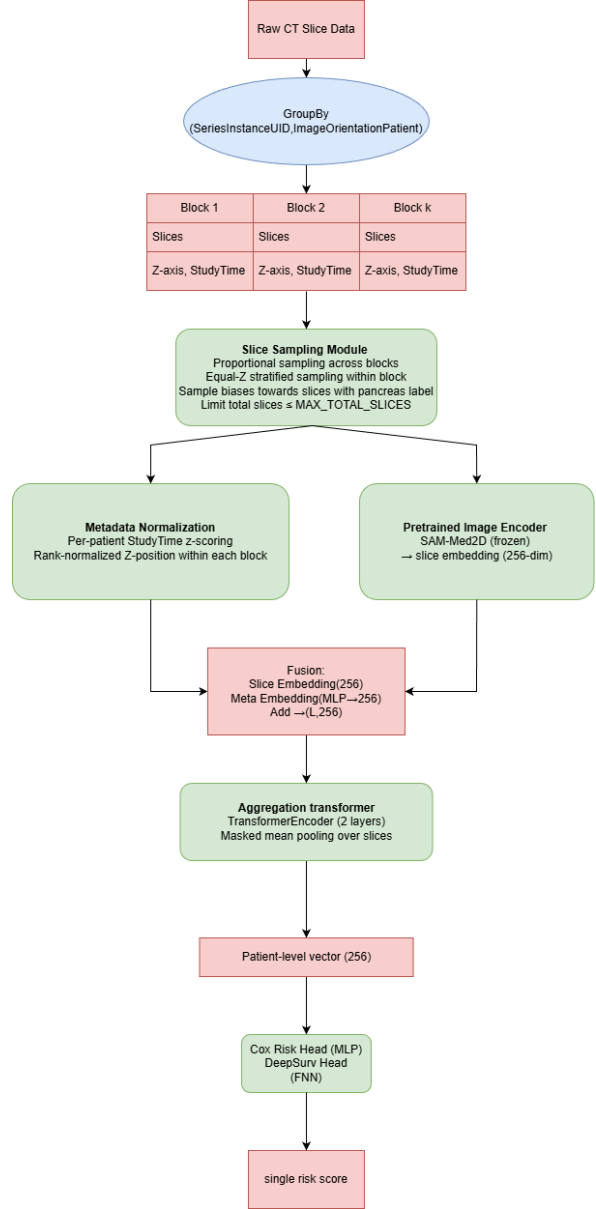


Figure 1: Overview of the image-processing pipeline. Raw CT slices are grouped into anatomical blocks, sampled, enriched with metadata, encoded using a pretrained SAM-Med2D model, and aggregated via a Transformer to produce a patient-level representation.

Metadata Normalization and Embedding.

Each slice is associated with acquisition metadata such as `StudyTime` and `Z-axis` position. We normalize `StudyTime` using per-patient z-scoring and convert `Z-position` into a rank-normalized coordinate within each block. These normalized

metadata features are passed through a lightweight MLP to produce a 256-dimensional metadata embedding:

$$m_i = \text{MLP}_{\text{meta}}(\text{meta}_i) \in R^{256}.$$

Pretrained Slice Encoder (SAM-Med2D). Image features are extracted using a frozen SAM-Med2D encoder:

$$z_i = f_{\text{SAM-Med2D}}(x_i), \quad z_i \in R^{256},$$

where SAM-Med2D provides slice-level representations informed by large-scale medical pretraining.

Slice-Level Fusion. For each slice, the image embedding and metadata embedding are combined by elementwise addition:

$$\tilde{z}_i = z_i + m_i,$$

yielding a metadata-informed slice representation of dimension 256. The resulting slice sequence is

$$\tilde{Z} = \{\tilde{z}_1, \tilde{z}_2, \dots, \tilde{z}_L\},$$

where $L \leq \text{MAX_TOTAL_SLICES}$.

Aggregation Transformer. To capture long-range anatomical dependencies across slices, we apply a two-layer TransformerEncoder to $\tilde{Z}$. The transformer produces contextualized slice embeddings, which are aggregated into a study-level representation using masked mean pooling:

$$z_{\text{patient}} = \text{MeanPool}(\text{Transformer}(\tilde{Z})).$$

This yields a fixed 256-dimensional patient-level imaging vector regardless of study length.

Output Representation. The patient-level vector is passed to the Cox projection head for time-to-event prediction.

3.4 Clinical Feature Encoder

Structured clinical variables (demographics, laboratory tests, encounter history, and behavior indicators) are processed through a two-layer multilayer perceptron (MLP):

$$h_{\text{clin}} = \text{MLP}_{\theta}(x_{\text{clin}}),$$

where the MLP consists of:

Linear $\rightarrow$ ReLU $\rightarrow$ Dropout $\rightarrow$ Linear $\rightarrow$ ReLU.

The output dimension is fixed at 128.

3.5 Multimodal Fusion

Imaging and clinical representations are fused by concatenation:

$$h = [z_{\text{vol}}; h_{\text{clin}}].$$

This fused feature is passed through a projection head:

$$r = g_{\phi}(h),$$

which outputs a scalar log-risk score r .

3.6 Cox Partial-Likelihood Objective

We optimize the model using the Cox proportional hazards loss. Given survival durations t_i and event indicators δ_i , the loss is:

$$\mathcal{L}_{\text{Cox}} = - \sum_{i:\delta_i=1} \left(r_i - \log \sum_{j \in \mathcal{R}(t_i)} e^{r_j} \right),$$

where $\mathcal{R}(t_i)$ is the risk set. We implement this loss directly in PyTorch, enabling end-to-end training with backpropagation.

3.7 Experimental Design and Evaluation

3.7.1 Prediction Task Definition

The prediction task is formulated as a time-to-event problem, where the objective is to estimate each patient’s risk of progressing to Chronic Pancreatitis (CP) following an initial Acute Pancreatitis (AP) episode. Models output a continuous risk score based on clinical features, imaging features, or their combination. Model performance is evaluated at discrete prediction horizons (1–5 years), reflecting clinically relevant time frames for CP progression.

3.7.2 Label Construction Strategies

We consider three label construction strategies.

- `classic_survival`: Uses the original time-to-event formulation, where the duration is the observed CP event time if progression occurs and otherwise the last follow-up time; the event indicator is given by the recorded CP diagnosis. No prediction horizon is applied.
- `fixed_window_survival`: Defines a horizon-specific endpoint by setting the event indicator to 1 only if CP occurs within the chosen horizon, and clipping all durations to be at most the horizon.

- `optimistic_window_survival`: Uses the same within-horizon event definition as `fixed_window_survival`, but assigns all censored cases a duration equal to the horizon, effectively treating early-censored patients as CP-free through the horizon.

3.7.3 Definition of Meaningful Datapoints

For a given prediction horizon, we define a *meaningful datapoint* as a patient whose Chronic Pancreatitis (CP) event status at that horizon can be determined without ambiguity. Concretely, a datapoint is considered meaningful if it satisfies one of the following conditions:

- The patient develops CP at or before the evaluation horizon.
- The patient develops CP after the evaluation horizon.
- The patient does not develop CP and has follow-up extending beyond the evaluation horizon.

Patients who are censored prior to the evaluation horizon without a CP event are excluded, as their event status at the horizon cannot be determined.

3.7.4 Data Inclusion Strategies

Experiments are conducted under two data inclusion settings:

- **Controlled**: Only meaningful datapoints are included, excluding patients who are censored prior to the evaluation horizon.
- **Uncontrolled**: All available datapoints are included, comprising both meaningful and ambiguous cases.

Under the controlled setting, labeling strategies `fixed_window_survival` and `optimistic_window_survival` produce identical labels and durations by construction. Consequently, redundant model training using `optimistic_window_survival` under the controlled setting is omitted.

Because evaluation is restricted to meaningful datapoints, both controlled and uncontrolled settings share an identical set of evaluation samples at each horizon, enabling direct comparison of model performance across settings.

Table 1: Distribution of meaningful datapoints under the 5-year prediction horizon.

Split	Meaningful	Total
Train	312	1213
Validation	104	405
Test	96	405

3.7.5 Dataset Splits

The dataset is partitioned at the patient level into training, validation, and test sets using a fixed stratified split (60%/20%/20%), where stratification is performed jointly on raw event status and discretized time-to-event to preserve outcome prevalence across splits.

As shown in Table 1, the stratified split implicitly preserves a comparable 6:2:2 proportion of meaningful datapoints across training, validation, and test sets.

The set of meaningful datapoints is intentionally held constant across controlled and uncontrolled settings at each prediction horizon. Consequently, models trained under different settings are evaluated on identical meaningful samples, enabling direct and fair comparison.

To prevent data leakage across modalities, the dataset split is aligned with the imaging pipeline: imaging representations are learned using only the training and validation sets, and the resulting fixed embeddings are subsequently used for downstream model training and evaluation, with the test set strictly held out during image representation learning.

3.7.6 Models training

We experimented with six models spanning clinical-only, imaging-only, and multimodal settings:

- **Clinical survival models**: a Cox proportional hazards model implemented with the `lifelines` package (CoxPH), a linear Cox model implemented in PyTorch (LinearCox), and a MLP model using clinical features only (MLP-Clinical).
- **Clinical classification baseline**: an XGBoost classifier trained to predict horizon-specific CP occurrence.
- **Imaging-only survival model**: a survival model trained on fixed image embeddings extracted from the imaging pipeline.

- **Multimodal survival model:** a fusion model combining clinical and imaging representations, trained using the same Cox partial-likelihood objective.

Model training is conducted over all combinations of prediction horizon (1–5 years), label construction strategy, and data inclusion strategy. For each configuration, a horizon-specific dataset is constructed according to the specified label method and controlled or uncontrolled setting, and models are trained independently on the resulting data.

Clinical survival models (CoxPH, LinearCox, and MLP-Clinical), imaging-only survival models, and multimodal fusion models are trained using a Cox partial-likelihood objective. Neural network-based models are trained with multiple random initializations to account for optimization variability, while CoxPH and XGBoost baselines are trained once per configuration. Certain redundant or ill-defined configurations are omitted, including optimistic window labeling under the controlled setting and XGBoost trained with classic survival labels, as these yield equivalent or invalid training objectives by construction.

3.7.7 Evaluation Metric and Horizon Compatibility

Model performance is evaluated using the area under the receiver operating characteristic curve (AUC), computed on meaningful datapoints only. AUC is chosen because it measures discriminative ability independent of decision thresholds and is robust to class imbalance, which varies substantially across evaluation horizons.

For a given evaluation horizon t , AUC measures a model’s ability to rank patients who develop Chronic Pancreatitis (CP) within t years above those who do not. Survival model outputs are converted into horizon-specific event probabilities, while classification models directly produce horizon-aligned prediction probability.

Importantly, models trained with a fixed training horizon n are evaluated only at horizons $t \leq n$. This constraint ensures that evaluation is consistent with the label construction used during training and avoids extrapolating predictions beyond the time range for which supervision is available. As a result, a model trained with a 3-year horizon contributes to AUC evaluation at 1–3 years, but not at 4–5 years.

Consequently, AUC values are not comparable across evaluation horizons (e.g., AUC@1y vs.

AUC@3y) due to horizon-dependent changes in the set of meaningful datapoints; in contrast, for any fixed evaluation horizon t , AUC provides a valid basis for comparing different models and training configurations, as long as the evaluation respects the constraint $t \leq n$.

3.8 Model Output

Survival-based model produces:

1. a scalar risk score r , where larger values indicate higher risk of CP progression;
2. an estimated survival curve $S(t | X)$ derived from the Cox formulation.

Classification-based model (XGBoost) outputs a scalar prediction score representing the likelihood of CP occurrence under the corresponding horizon-specific label construction.

4 Results and Analysis

4.1 Effect of Training Horizon on Risk Discrimination

As illustrated in Appendix A.1, we observe that models trained with larger horizons tend to achieve larger AUC (discriminative performance) when evaluated at shorter horizons. The only exception occurs under the uncontrolled data inclusion strategy with `classic_survival` labeling, where performance remains invariant to the training horizon due to identical labels and durations across configurations.

This pattern is explained by the distribution of meaningful datapoints. Short-horizon training exhibits severe class imbalance, with relatively few positive CP events, resulting in weak supervision and unstable risk ranking. In contrast, longer horizons substantially increase the number of positive cases and yield a more balanced training signal, enabling models to learn more robust and generalizable risk patterns. Because longer-horizon positives subsume early CP events, models trained at larger horizons generalize well when evaluated at shorter horizons.

4.2 Effect of Data Inclusion Strategy on Risk Discrimination

Across most training horizons, experiment configurations, and model families, models trained under the controlled data inclusion strategy consistently outperform their uncontrolled counterparts in

Table 2: Distribution of meaningful and ambiguous datapoints across prediction horizons.

Horizon	Pos	Neg	Ambig	Total
1y	67	821	1135	2023
2y	144	576	1303	2023
3y	174	453	1396	2023
4y	196	371	1456	2023
5y	219	293	1511	2023

terms of AUC, with occasional deviations at the 1-year horizon for the classification-based XGBoost model. Example of AUC values at the 2-year horizon for both data inclusion strategies are reported in Appendix A.2.

The performance gap arises from how ambiguous datapoints are handled during training. Controlled models are trained exclusively on meaningful datapoints, i.e., patients whose CP status at the prediction horizon is unambiguous. In contrast, uncontrolled models include all available datapoints, including a large number of ambiguous cases with insufficient follow-up. Although such datapoints may contain partial information (e.g., a patient remains CP-free for a limited time), they are labeled as non-events under horizon-based labeling and can introduce false-negative noise if CP occurs later within the horizon but is unobserved.

Table 2 shows that ambiguous datapoints dominate the dataset, their inclusion overwhelms the signal from meaningful cases and distorts the learned risk patterns. By restricting training to meaningful datapoints only, controlled models are exposed to cleaner supervision and are able to learn more reliable discriminative structures for CP progression.

Overall, these results indicate that while additional data may appear useful, it is not necessarily informative for horizon-specific risk prediction. In contrast, meaningful datapoints consistently provide reliable supervision, leading to improved and more stable model performance.

4.3 Effect of Label Construction on Horizon-Specific Discrimination

Under the uncontrolled data inclusion strategy, we observe a clear horizon-dependent contrast between `fixed_window_survival` and `optimistic_window_survival`. At the 1-year evaluation horizon, optimistic-window labeling generally outperforms fixed-window labeling, whereas at longer horizons (2–5 years), fixed-window labeling consistently yields superior dis-

Table 3: Comparison of `fixed_window_survival` and `optimistic_window_survival` under uncontrolled settings. Models are trained with a 5-year horizon and evaluated at 1-year and 2-year horizons.

Model	AUC (1-year)		AUC (2-year)	
	Fixed	Optim	Fixed	Optim
CoxPH	0.6080	0.5900	0.6910	0.5780
LinearCox	0.6065	0.6165	0.6875	0.5845
MLP	0.5585	0.6020	0.6555	0.5650
IMAGE	0.6975	0.6910	0.6485	0.6480
FUSION	0.6230	0.6810	0.6860	0.6245

crimination (Table 3).

This behavior is explained by how the two label constructions handle censor-time information. Optimistic-window labeling collapses censoring times by assigning nearly all negative cases a duration equal to the horizon, thereby removing censor-time heterogeneity. In contrast, fixed-window labeling preserves the original follow-up durations of censored cases, producing a wide and continuous distribution of censor times.

At short evaluation horizons, such as 1 year, most censor-time variability either lies beyond the evaluation window or is concentrated within a narrow range below 1 year. In this regime, differences in follow-up time (e.g., 0.2 versus 0.8 years) are weakly aligned with the binary objective of predicting CP occurrence within one year and therefore contribute limited ranking signal. Preserving this heterogeneity can instead introduce noise, explaining the diminished or reversed performance of fixed-window labeling at AUC@1y.

At longer evaluation horizons, censor-time heterogeneity becomes directly informative. Differences such as being censored at 1 versus 3 years meaningfully reflect differential risk of remaining CP-free within a 2–5 year window. Fixed-window labeling allows survival models to exploit this ordering, while optimistic-window labeling eliminates it entirely by construction. As a result, the benefit of preserving censor-time heterogeneity becomes increasingly pronounced at mid- and long-term horizons.

4.4 Effect of Classic Survival Labeling on Cross-Horizon Generalization

Across all evaluation horizons (1–5 years), models trained with `classic_survival` consistently outperform those trained with `fixed_window_survival` under both con-

trolled and uncontrolled data inclusion strategies.

This advantage arises because classic survival preserves the full time-to-event information, including late CP events and fine-grained censoring heterogeneity. In contrast, fixed-window labeling truncates follow-up time at the horizon and reclassifies late CP events as censored negatives, discarding informative risk signal.

Although fixed-window labeling more closely matches the downstream task definition of predicting CP occurrence within a fixed time window, it sacrifices statistical efficiency by collapsing heterogeneous outcomes. Classic survival instead learns a global risk ordering that generalizes more effectively across evaluation horizons.

4.5 Limited Sensitivity of Image-Only Models to Training Horizon

We observe that increasing the training horizon yields only marginal improvements for the image-only model. This behavior is consistent across evaluation horizons, as illustrated by the representative results at the 1-year evaluation horizon (Section A.1).

Increasing the training horizon modifies the label definition (e.g., predicting CP within 2 years versus within 5 years), but does not introduce additional information into the image features themselves. Although image embeddings are extracted from multiple CT scans with relative temporal context, they primarily encode the patient’s current disease state and accumulated structural severity, rather than explicit information about the remaining time until CP onset. Consequently, extending the training horizon alters the supervision signal without providing new feature structure that image-only models can effectively exploit.

4.6 Model Family Comparison and Class Imbalance Effects

Across evaluation horizons, clinical-based survival models consistently achieve strong discrimination, with CoxPH performing best, followed by LinearCox and MLP. This ordering reflects the advantage of explicitly modeling time-to-event structure in clinical features, particularly under controlled settings where labels are unambiguous.

XGBoost exhibits increasing AUC at longer evaluation horizons, consistent with its strength in binary window-based classification when more long-term outcomes are observable and class balance improves. However, its performance is less stable

Table 4: Distribution of meaningful true positives and true negatives under controlled experiments across evaluation horizons.

Horizon	Split	TP	TN	Total
1y	Train	41	495	536
1y	Val	13	163	176
1y	Test	13	163	176
3y	Train	102	278	380
3y	Val	34	90	124
3y	Test	38	85	123
5y	Train	128	184	312
5y	Val	45	59	104
5y	Test	46	50	96

Table 5: AUC across evaluation horizons under the best-performing configuration (classic_survival, controlled, training horizon = 5 years).

Model	Evaluation Horizon (years)				
	1y	2y	3y	4y	5y
IMAGE	0.706	0.667	0.607	0.626	0.595
MLP	0.652	0.734	0.776	0.814	0.809
FUSION	0.704	0.778	0.806	0.843	0.823

at short horizons, where severe class imbalance limits learnable signal.

In contrast, the fusion model maintains robust performance across all horizons by integrating complementary information from imaging and clinical features, rather than relying on a single modality. This allows fusion to remain competitive even when one modality is disadvantaged by horizon-specific characteristics.

Importantly, apparent performance differences across evaluation horizons are partly driven by substantial shifts in class balance. Early horizons are characterized by extreme imbalance between positive and negative cases, which increases variance and weakens supervision, while longer horizons exhibit more balanced outcome distributions that support more stable risk discrimination.

4.7 Fusion Benefits Across Evaluation Horizons

Under the best-performing configuration (classic_survival, controlled, training horizon = 5 years), the relative strengths of the image-only and clinical-only models vary across evaluation horizons (Table 5). **Importantly, performance comparisons are made within each evaluation horizon, rather than across model**

training horizons.

The image-only model performs relatively well at short horizons, while its discriminative ability weakens at mid- to long-term horizons. In contrast, the clinical MLP exhibits stronger performance at mid- and long-term horizons, reflecting its ability to capture longitudinal risk factors that are more predictive over extended time frames. These trends do not imply that performance degrades or improves across horizons in absolute terms, but rather that the relative ranking of modalities differs within each horizon-specific evaluation.

Across all evaluation horizons, the fusion model consistently matches or exceeds the stronger of the two individual modalities at the same horizon. When imaging features provide greater short-term signal, fusion maintains strong discrimination; when clinical features dominate at longer horizons, fusion continues to perform well. This behavior suggests that the fusion model effectively leverages complementary information from imaging and clinical features in a horizon-dependent manner, yielding robust performance without reliance on a single modality.

4.8 Summary of Empirical Findings

Our experimental analysis yields three consistent findings across model families and evaluation horizons.

First, `classic_survival` labeling provides the strongest overall risk modeling performance. By preserving the full time-to-event structure, including late CP events and censoring heterogeneity, classic survival consistently achieves higher discrimination than window-based labeling strategies. In contrast, `fixed_window_survival` and `optimistic_window_survival`, while more closely aligned with horizon-specific classification objectives, discard informative temporal signal and exhibit weaker empirical performance.

Second, controlled setting for data inclusion strategy consistently outperform uncontrolled settings. Restricting training to meaningful datapoints removes ambiguous cases with insufficient follow-up, allowing models to learn clearer and more reliable risk patterns.

Third, larger training horizons lead to improved learning when combined with controlled data inclusion. Using a 5-year training horizon ensures that all retained patients have unambiguous outcome status relative to the horizon, producing a clean and well-aligned training set for evaluation across 1–5

year horizons.

4.9 Best-Performing Model Configuration

Although certain single-modality models achieve strong performance at specific horizons—such as the image-only model at the 1-year horizon and XGBoost at the 5-year horizon—these gains do not generalize consistently across evaluation horizons. In contrast, the fusion model demonstrates stable and strong discrimination throughout the entire 1–5 year range.

Based on these results, we select the following configuration as the overall best-performing model:

- Model: `FUSION`
- Labeling: `classic_survival`
- Data inclusion: `controlled`
- Training horizon: 5 years

Under this configuration, the fusion model achieves consistently high AUC across all evaluation horizons (Figure 3 in Appendix A.3). And the fusion model achieves the following performance across evaluation horizons, reported as the first quartile (Q1), median, and third quartile (Q3) over 10 independent training runs:

- 1-year AUC: (0.688, 0.704, 0.717)
- 2-year AUC: (0.762, 0.778, 0.780)
- 3-year AUC: (0.789, 0.806, 0.812)
- 4-year AUC: (0.831, 0.843, 0.853)
- 5-year AUC: (0.817, 0.823, 0.834)

These results indicate that multimodal fusion effectively integrates complementary imaging and clinical signals, yielding robust risk discrimination without reliance on a single modality.

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A Additional Experimental Results

A.1 AUC at the 1-Year Horizon

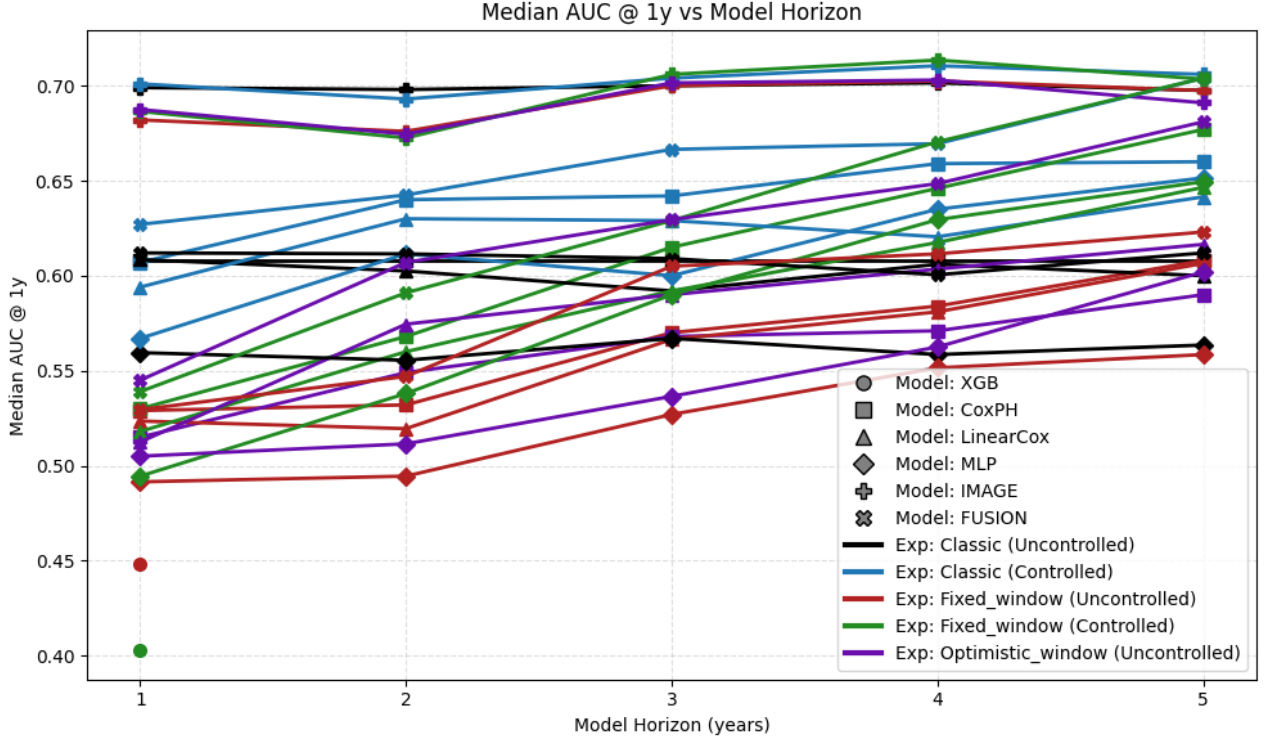


Figure 2: AUC evaluated at the 1-year horizon for all models under different training horizon configurations.

A.2 Data Inclusion Strategy Comparison at the 2-Year Horizon

Table 6: AUC evaluated at the 2-year horizon under different label construction and data inclusion strategies.

Model	Classic-Uncontrolled	Classic-Controlled	Fixed-Uncontrolled	Fixed-Controlled
CoxPH	0.7040	0.7135	0.6640	0.6800
LinearCox	0.6935	0.6975	0.6755	0.6815
MLP	0.6675	0.6900	0.6515	0.6845
IMAGE	0.6635	0.6655	0.6550	0.6580
FUSION	0.6910	0.7295	0.6760	0.7140
XGBoost	—	—	0.5270	0.6830

A.3 Model Comparison under Best Configuration

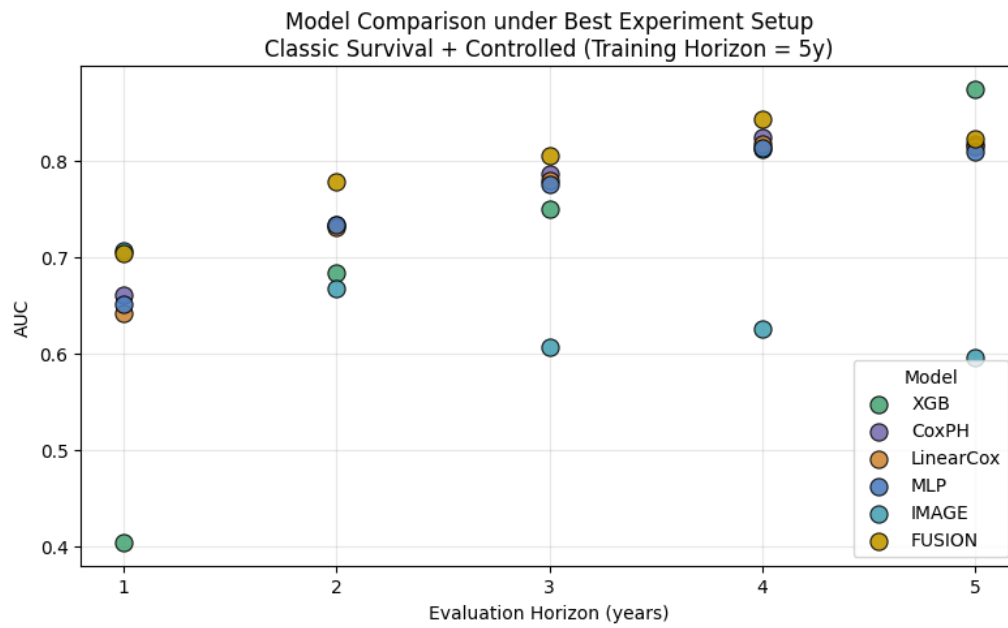


Figure 3: Model comparison under the best-performing configuration (`classic_survival`, `controlled`, training horizon = 5 years).